PROPER SUPPORT OF A FINITE TYPE MODULE IN THREE FORMULATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof identifies the underlying spaces of the two supports for (1)-(2), produces the factorization through the scheme-theoretic support for (2)-(3), and closes with the heredity of properness under closed subsets

Editorial note. The label lemma-module-support-proper-over-base names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the support of a finite type quasi-coherent module is closed (Morphisms of Spaces, Lemma spaces-morphisms-lemma-support-finite-type)
- the definition of properness over the base for closed subsets (definition-proper-over-base of the same chapter)
- the scheme-theoretic support of F is a closed subspace with underlying topological space Supp(F) (Morphisms of Spaces, Definition spaces-morphisms-definition-scheme-theoretic-support)
- closed subsets of subsets proper over the base remain proper (lemma-closed-closed-proper-over-base of the same chapter)

Source reference. Original reference 0CYM: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1086–1086. Prerequisite pointer invoked at proof line(s) 1270; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CYN: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1103–1103. Prerequisite pointer invoked at proof line(s) 1280; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 07U1: sources/stacks-chow/tags.txt, source lines 9956–9956. Prerequisite pointer invoked at proof line(s) 1257; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07TZ: sources/stacks-chow/tags.txt, source lines 9954–9954. Prerequisite pointer invoked at proof line(s) 1269; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite type. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_X$-module. The following are equivalent*

(1) *the support of $\mathcal{F}$ is proper over Y,*
(2) *the scheme theoretic support of $\mathcal{F}$ (Morphisms of Spaces, Definition 07U1) is proper over Y, and*
(3) *there exists a closed subspace $Z \subset X$ and a finite type, quasi-coherent $\mathcal{O}_Z$-module $\mathcal{G}$ such that (a) $Z \to Y$ is proper, and (b) $(Z \to X)_*\mathcal{G} = \mathcal{F}$.*

Proof. The support $\mathrm{Supp}(\mathcal{F})$ of $\mathcal{F}$ is a closed subset of $|X|$, see Morphisms of Spaces, Lemma 07TZ. Hence we can apply Definition 0CYM. Since the scheme theoretic support of $\mathcal{F}$ is a closed subspace whose underlying closed subset is $\mathrm{Supp}(\mathcal{F})$ we see that (1) and (2) are equivalent by Definition 0CYM. It is clear that (2) implies (3). Conversely, if (3) is true, then $\mathrm{Supp}(\mathcal{F}) \subset |Z|$ and hence $\mathrm{Supp}(\mathcal{F})$ is proper over Y for example by Lemma 0CYN. □